

Homework for Week 13

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May 29, 2026

1.

Proof. 可以将 u 按 $\{\Phi_k\}$ 展开, 得到 $u = \sum_k c_k \Phi_k$, 带回 u 满足的偏微分方程即得到

$$\sum_k \lambda_k c_k \Phi_k = f \Rightarrow \lambda_k c_k = A_k,$$

故

$$u = \sum_k \frac{A_k}{\lambda_k} \Phi_k.$$

□

2. 对于本题中的方程及齐次条件, 可以写出相应的本征值和本征函数为

$$\lambda_{mn} = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2$$
$$w_{mn}(x, y) = \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$$

将 $u(x, y)$ 按 $w_{mn}(x, y)$ 展开

$$u(x, y) = \sum_{m=1}^{\infty} \sum_{n=0}^{\infty} c_{mn} \sin \frac{m\pi x}{a} \cos \frac{n\pi y}{b}$$

并代入原定解问题的偏微分方程中得到

$$\left[\left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2\right] c_{mn} = \int_0^b \cos \frac{n\pi y}{b} dy \int_0^a \sin \frac{m\pi x}{a} f(x, y) dx$$

求解该代数方程即得到 $u(x, y)$ 的展开系数.

3. 要求积分核为 θ 对应算符的本征函数, 则应当取为 Legendre 函数 $P_l(\cos \theta)$. 此时 u 的像函数为

$$U_l(r, \phi) = \int_0^\pi u(r, \theta, \phi) P_l(\cos \theta) \sin \theta \, d\theta,$$

满足定解问题

$$\begin{aligned} \frac{d}{dr} \left(r^2 \frac{dU_l}{dr} \right) - l(l+1)U_l(r) &= 0 \\ U_l(1) &= \int_{-1}^1 f(x) P_l(x) \, dx \end{aligned}$$

一般不能对球内的 r 进行积分变换, 积分区间不符同时也仅有边界条件 $r = 1$ 处的 Dirichlet 边界条件, 在 $r = 0$ 处仅有有界性条件.

4. 可以将题述问题写为

$$\begin{cases} \frac{\partial u}{\partial t} - \kappa \frac{\partial^2 u}{\partial x^2} = 0, & -\infty < x < \infty, t > 0 \\ u|_{x \rightarrow \pm\infty} < \infty, & t > 0 \\ u|_{t=0} = u_0 H(x), & -\infty < x < \infty \end{cases}$$

作 Laplace 变换, 令

$$u(x, t) \doteq U(x, p) = \int_0^\infty u(x, t) e^{-pt} \, dt,$$

则定解问题化为

$$pU(x, p) - \kappa \frac{d^2 U(x, p)}{dx^2} = u_0 H(x),$$

由此得到

$$U(x, p) = \frac{1}{2} \frac{1}{\sqrt{\kappa p}} \int_{-\infty}^\infty u_0 H(x') \exp \left[-\sqrt{\frac{p}{\kappa}} |x - x'| \right] dx' = \begin{cases} \frac{u_0}{2p} \exp \left(\sqrt{\frac{p}{\kappa}} x \right), & x < 0, \\ \frac{u_0}{2p} [2 - \exp(-\sqrt{\frac{p}{\kappa}} x)], & x \geq 0, \end{cases}$$

所以

$$u(x, t) = \begin{cases} \frac{u_0}{2} \operatorname{erfc} \left(-\frac{x}{2\sqrt{\kappa t}} \right), & x < 0, \\ u_0 \left(1 - \frac{1}{2} \operatorname{erfc} \frac{x}{2\sqrt{\kappa t}} \right), & x \geq 0. \end{cases}$$

5. 对时间变量作 Laplace 变换, 定解问题化为

$$p^2 U(x, p) - a^2 \frac{d^2 U(x, p)}{dx^2} = \frac{p}{p^2 + \omega^2}, \quad 0 < x < \infty$$

$$U(0, p) = 0, \quad \left. \frac{dU}{dx} \right|_{x \rightarrow \infty} = 0$$

结合边界条件得到其解为

$$U(x, p) = \frac{1}{p} \frac{1}{p^2 + \omega^2} (1 - e^{-px/a}),$$

逆变换得到原定解问题的解为

$$u(x, t) = \frac{1}{\omega^2} (1 - \cos \omega t) - \frac{1}{\omega^2} \left[1 - \cos \omega \left(t - \frac{x}{a} \right) \right] H \left(t - \frac{x}{a} \right).$$

6. 首先求解定解问题

$$\frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial u}{\partial \rho} \right) + \frac{\partial^2 u}{\partial z^2} = 0, \quad 0 < \rho < a, -\infty < z < \infty$$

$$u \Big|_{\rho=a} = V_0 e^{-\kappa|z|} \operatorname{sgn} z, \quad -\infty < z < \infty$$

对 z 作 Fourier 变换得到关于 $U(\rho, k)$ 的定解问题

$$\frac{1}{\rho} \frac{d}{d\rho} \left[\rho \frac{dU(\rho, k)}{d\rho} \right] - k^2 U(\rho, k) = 0, \quad 0 < \rho < a$$

$$U(a, k) = -\frac{V_0}{\sqrt{2\pi}} \frac{2ik}{\kappa^2 + k^2}$$

上述方程为 0 阶虚宗量 Bessel 方程, 结合边界条件得到其解为

$$U(\rho, k) = -\frac{V_0}{\sqrt{2\pi}} \frac{2ik}{\kappa^2 + k^2} \frac{I_0(k\rho)}{I_0(ka)},$$

逆变换并取 $\kappa \rightarrow 0$ 得到原定解问题解为

$$u(\rho, z) = \frac{2V_0}{\pi} \int_0^\infty \frac{I_0(k\rho)}{I_0(ka)} \frac{\sin kz}{k} dk.$$